Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.1 High-Frequency Transistor Model

Frequency Response

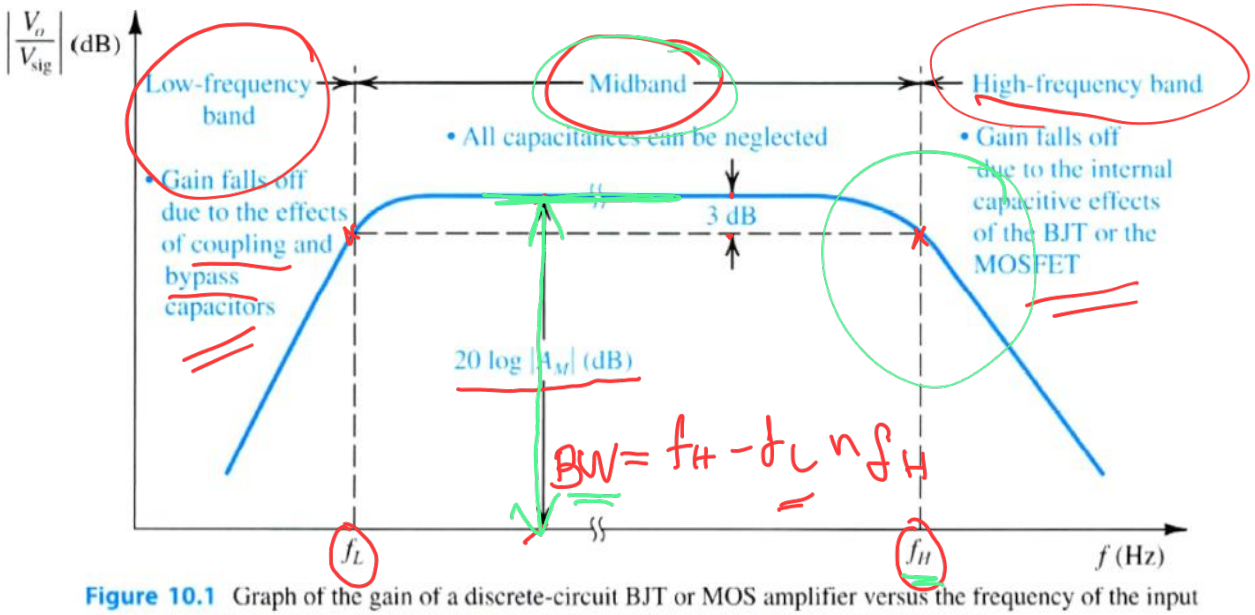


Figure 10.1 Graph of the gain of a discrete-circuit BJT or MOS amplifier versus the frequency of the input signal, showing the three frequency bands relevant to frequency-response determination.

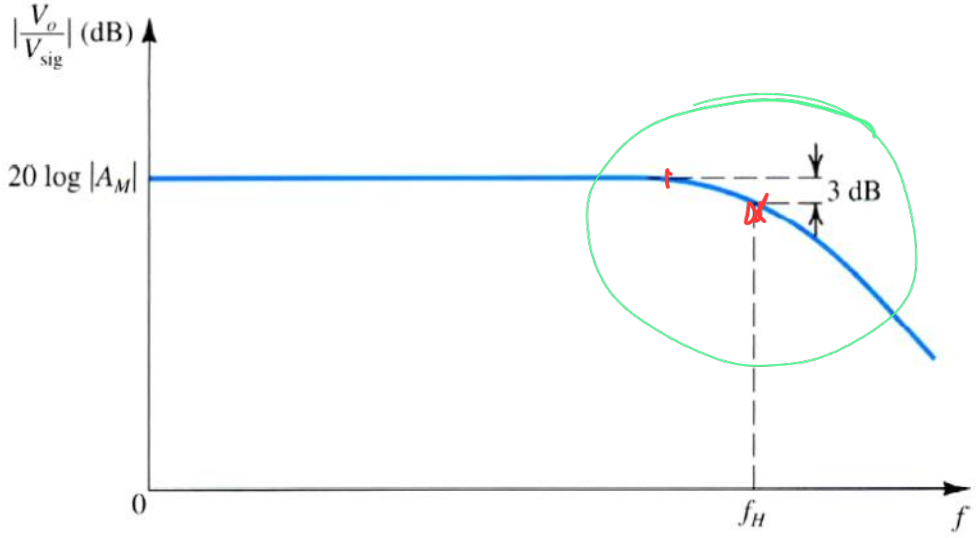


Figure 10.2 Frequency response of a direct-coupled (dc) amplifier. Notice that the gain does *not* fall off at low frequencies, and the midband gain A_M extends down to zero frequency.

Capacitances of a MOSFET

$$C_{ov} = WL_{ov}C_{ox} \qquad (10.1)$$

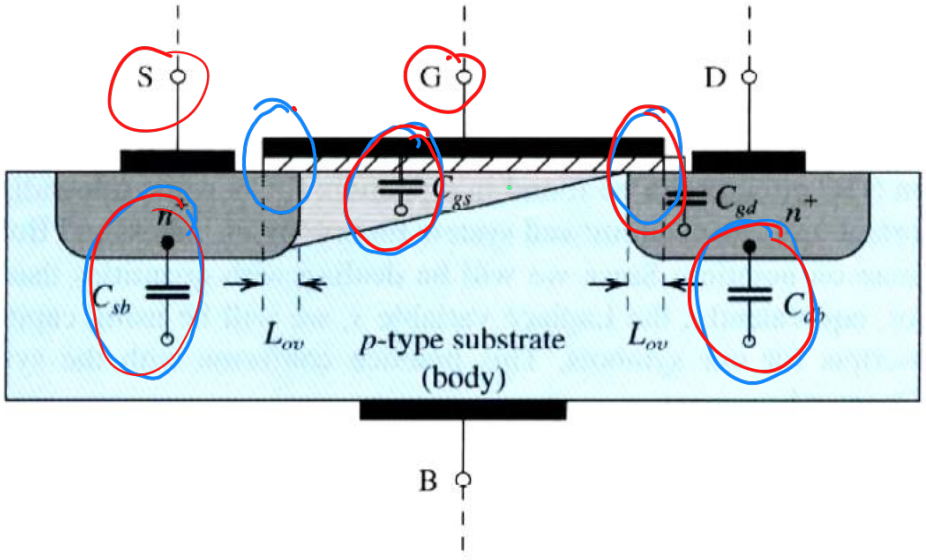


Figure 10.3 Cross section of an *n*-channel MOSFET operating in the saturation region. The four internal capacitances, C_{gs}, C_{gd}, C_{sb}, and C_{db}, are indicated. Note that the bias voltages are not shown. Also not shown, to keep the diagram simple, is the depletion region.

$$\text{Sat.} \quad \frac{2}{3}WLC_{ox} + C_{ov} = C_{gs}$$

$$C_{gd} = C_{ov}$$

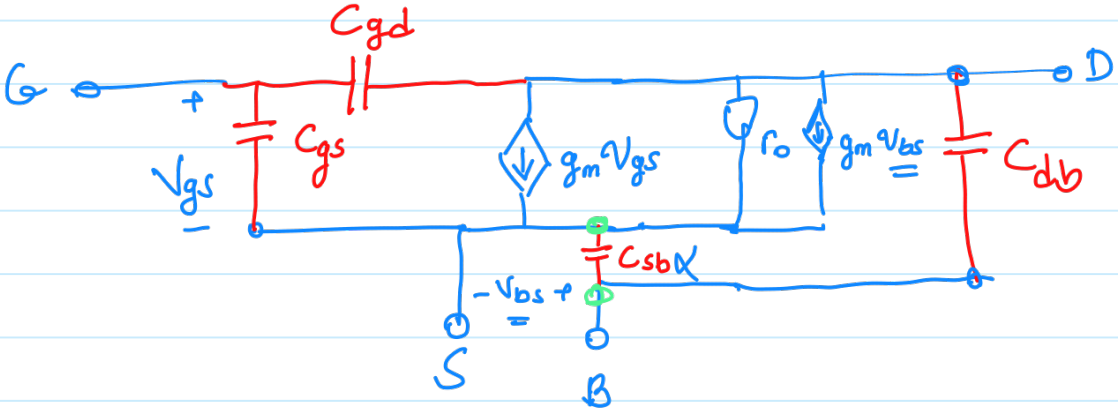
$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{V_{SB}}{V_0}}} \qquad C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{V_{DB}}{V_0}}}$$

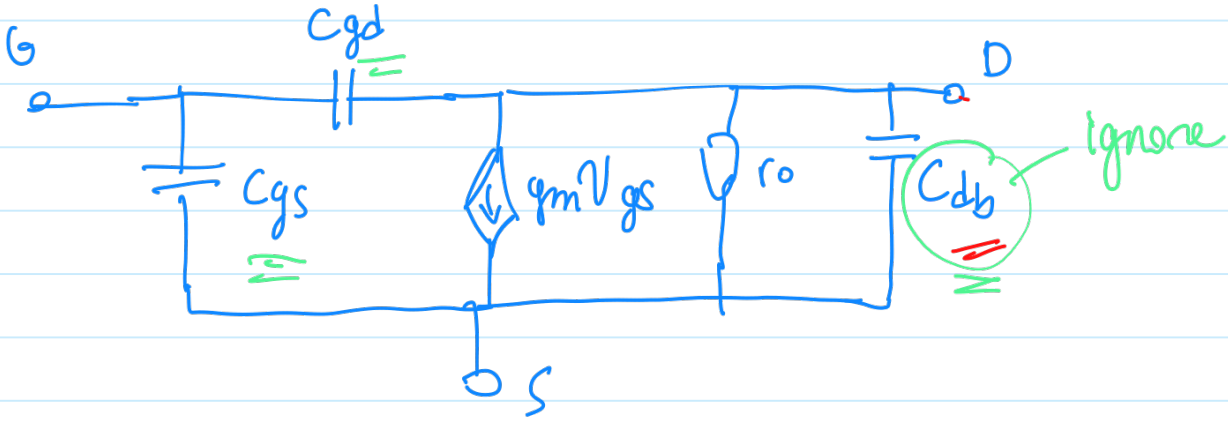
EXERCISE

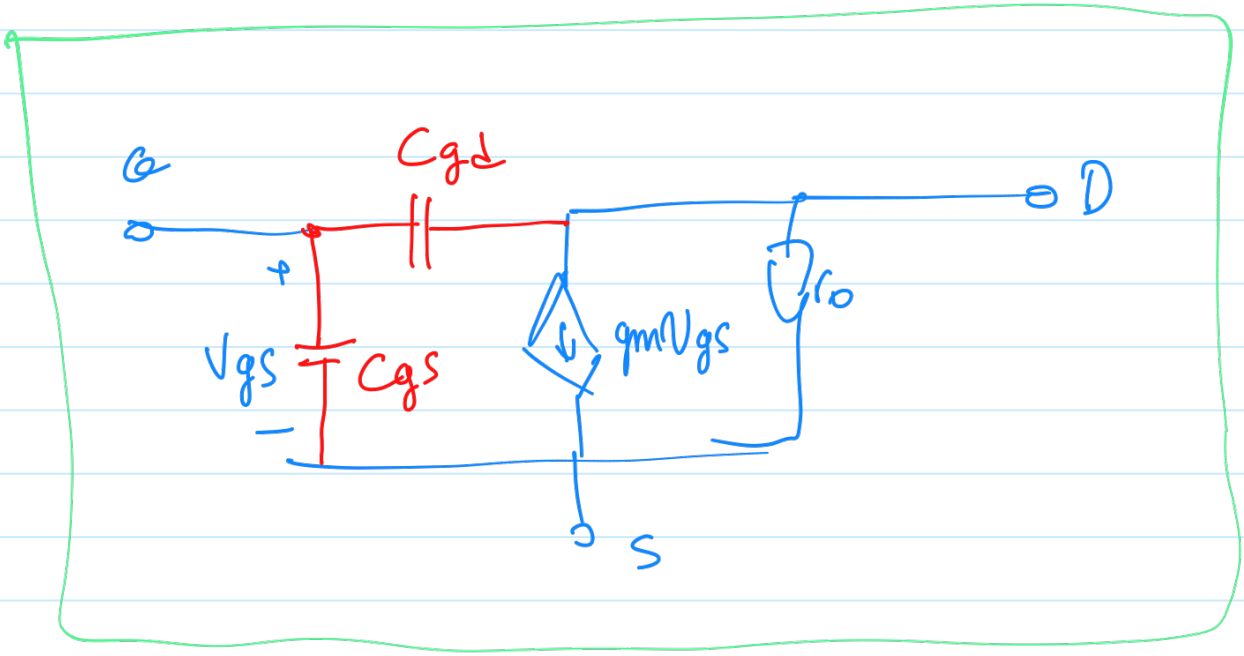
10.1 For an *n*-channel MOSFET with $t_{ox} = 3$ nm, $L = 0.15\,\mu$m, $W = 1.5\,\mu$m, $L_{ov} = 0.03\,\mu$m, $C_{sb0} = C_{db0} = 1$ fF, $V_0 = 0.9$ V, $V_{SB} = 0.2$ V, and $V_{DS} = 0.8$ V, calculate the following capacitances when the transistor is operating in saturation: C_{ox}, C_{ov}, C_{gs}, C_{gd}, C_{sb}, and C_{db}.

Ans. 11.5 fF/μm^2; 0.52 fF; 2.25 fF; 0.52 fF; 0.9 fF; 0.7 fF

The High Frequency Model of MOSFET





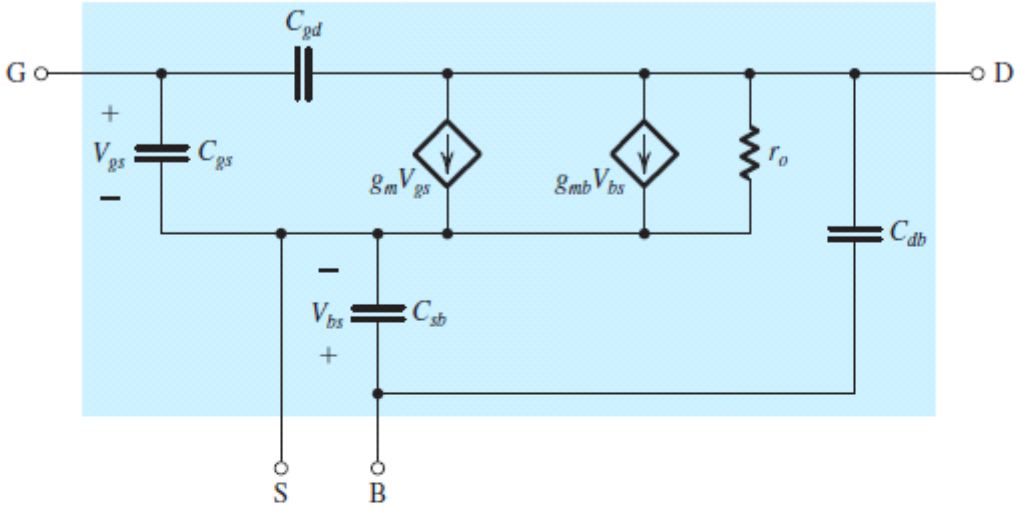


The MOSFET Unity Gain Frequency

Summary Table

Table 10.1 The MOSFET High-Frequency Model

Model



Model Parameters

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}| = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{|V_{OV}|}$$

$$g_{mb} = \chi g_m, \quad \chi = 0.1 \text{ to } 0.2$$

$$r_o = |V_A| / I_D$$

$$C_{gs} = \frac{2}{3} WLC_{ox} + WL_{ov} C_{ox}$$

$$C_{gd} = WL_{ov} C_{ox}$$

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{|V_{SB}|}{V_0}}}$$

$$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{|V_{DB}|}{V_0}}}$$

$$f_T = \frac{g_m}{2\pi (C_{gs} + C_{gd})}$$

Template